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Design, Implementation, and Deployment of an IoT Based Smart Energy Management System

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ABSTRACT Growth of population and the inception of new devices every day comes with an incessant rise in energy consumption and has brought great challenges in terms of energy management at the consumer side. With the evolution of technology, smart meters (SMs) are not only considered merely as tools to measure energy consumption but act as a main resource of energy management systems. The application of SM spans over a wide range of advantages, including accurate billing data, information of utilization at the user end, the establishment of two-way communication and remote control of the user equipment. SM is the most essential element of a smart power grid that with the help of any smart energy management system (SEMS), assesses, measures, controls, implements and communicates power allocation, utilization, and consumption at both, single device, and network level. Data provided by the SMs is used by power supply companies to revolutionize power distribution and consumption through various techniques such as, non-intrusive load monitoring and demand-side management (DSM). For efficient data gathering and utilization, internet of things (IoT) is emerging as a key partner in the power industry leading to effective resource management. This paper provides presentation, deployment, and validation of an IoT based SEMS strategy and its related benefits to overcome challenges of energy management at consumer side. The presented SEMS incorporates various communication interfaces and protocols to integrate with any software-based smart solution. In this work, Entrack software is used for data gathering and analysis. As a proof of concept, the presented system is tested and implemented in 4 different buildings of a well-known private company in Pakistan, i.e., Stylo Pvt. Ltd. The case study analysis in this work shows the effectiveness of the presented IoT based SEMS.

INDEX TERMS Energy Management System (EMS), Internet of Things (IoT), Smart Meter (SM), Smart Grid (SG)

I. INTRODUCTION

Exponential increase in urban development has created a challenging issue of energy consumption management in power sector. Especially, in the countries where temperatures are very high, devices like air conditioning consume massive power, creating even more challenging situation for the power sector to manage demand at consumer side. In recent years, a lot of smart solutions have been proposed to overcome the issue of demand side energy management. Such solutions include designing of energy management systems (EMS) using combination of various approaches like renewable resources (RES) based microgrids, smart grids, smart control of consumer devices using the local area networks (LANs) or internet of things (IoT) etc. With the inclusion of IoT in EMS has given birth to smart EMS (SEMS) as IoT enabled devices can be controlled smartly from anywhere in the world. In this work, we have presented the deployment and validation of an IoT based SEMS for providing efficient energy management. Our presented framework focuses on the part of monitoring and calculating the smart grid parameters using IoT enabled smart meters (SMs). To understand these terminologies, a brief overview of each is provided below.

A. SMART GRID

The smart grid (SG) is not only a set of smart energy meters involved in the power generation but is a group of various technologies and tools which allow integration, interaction, and control of all elements from the utility to consumer side for the smooth flow of power. The SG enables real-time monitoring of power system through visualization of all the necessary information and then acts accordingly for its smooth functioning, e.g., immediate balancing of demand and supply, bidirectional flow of electricity etc. [1][2][3].

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Smart grids are comprised of distributed generation (DG) source, (i.e., solar, wind and fossil fuel etc.) multiple loads (i.e., buildings, homes, industries), along with a datacenter which can handle the whole power grid. The infrastructure needs to satisfy the requirement of security, privacy, sustainability and reliability of the system [4]. The generic architecture of a smart grid with its key component is shown in Fig 1.

SG may be defined as a system that can effectively track, regulate, and protect electricity generation, transmission, and distribution systems. In this manner, it helps in energy management by observing and balancing the demand and supply curve. Which is very important because more often the power generation capacity does not fulfil the consumer's power demand. Furthermore, theft of electricity and shambolic handling of customer data also contributes to significant damages resulting in higher bills for the end-users [5][6].

A SG requires real and reliable information of demand-side power consumption to perform demand-side management (DSM). SMs in SGs assist consumer integration into the grid by offering two-way communication [7][8][9]. This ensures an effective delivery system for the utilities and more quality to the end-users. As the supply and demand balance needs real-time information exchange, so, modern and efficient technologies must be adopted to facilitate system integration and enable efficient decision-making [10].

B. SMART METER AND ITS IMPORTANCE

Technological literacy has led to responsible power utilization. Implying innovative technologies to power grids makes it more efficient. In this regard, it is vital to use the smart energy meters as it is the fundamental part of communication infrastructure in a SG [11][12]. SM, being a part of SG capacitates two-way communication in terms of command actuation of smart devices and information flow of power generation and consumption [13][14]. It records power usage and transmit data to the central data server periodically for the purpose of monitoring, control, and analysis. Furthermore, unlike normal meters, SMs are not just limited to measuring energy consumption but they can also measure a wide range of electrical variables for example current, voltage, frequency, power, and power factor. These electrical variables play a vital role in load management, fault analysis and load profiling. The choice of SM is driven by its additional functionalities and low cost. Using SMs with IoT open a lot of possibilities as this enables an SG to operate at a larger scale rather than a local scale due to the use of internet cloud services.

C. INTERNET OF THINGS (IoT)

The IoT is emerging as a key partner for power and resource management in intelligent system environments. It allows management and integration of equipment in the way of tracking, linking and responding to various applications [20][21]. It also enables two-way communication among

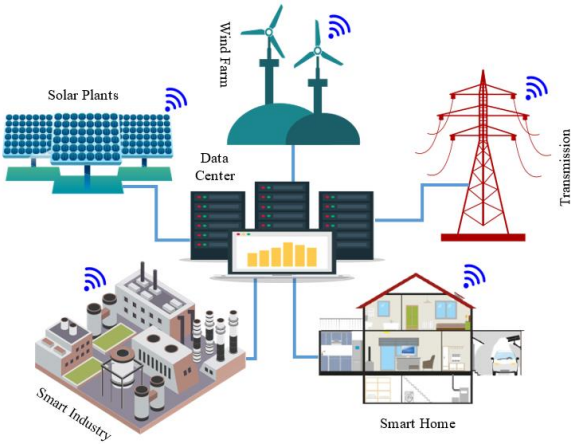


FIGURE 1. Smart Grid and its key components.

devices, networks and sensors with or without human intervention, which is crucial for a smart environment designed to effectively use resources and solve the associated challenges [22][23].

IoTs make it possible for internal equipment to communicate with the external environment and may influence the action taken. A SM system based on IoT mainly increases the SG's accuracy and productivity by enhancing its reliability. Collection and analysis of SG active devices data provides an opportunity of control to consumers and suppliers in terms of energy resource utilization in any desired manner. [24][25].

D. CONTRIBUTIONS

This work develops a SEMS design and implementation based on an IoT solution that incorporates SMs and an IoT middleware for efficient data analysis and management. The suggested solution is deployed at the consumer's end at 4 different locations of a well-known private industry in Pakistan, i.e., Stylo Pvt. Ltd, given in Table IV. The solution incorporates IoT operations and provides efficient monitoring of the supplied power. This paper's key contributions can be summed up as follows:

- The most critical environmental parameters where smart energy meters need to be deployed are identified and provided.
- Discuss the background, evolution, and types of meters along with functional features of the SMs.
- A proposal to address the dilemma of users' lack of knowledge on energy usage.
- System architecture, design, and integration of SMs including IoT middleware module.
- The demonstration, performance evaluation, and validation of the complete solution followed by a case study are provided.

The remainder of the paper is organized as follows. Section 2 presents the background and evolution of SEMS and SMs. The presented system architecture is described in Section 3. Demonstration, solution validation and case study are

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presented in section 4. The potential advantages of the presented system are discussed in section 5. Finally, the Conclusion and future directions are listed in section 6.

II. BACKGROUND AND EVOLUTION OF SMART ENERGY SYSTEMS AND SMART METERS

A. BACKGROUND

The idea of Electrical Grids was introduced in 1800s, but they were adopted in developed countries in 1960 as by that time the power grid had substantial penetration and capabilities, and adequate efficiency as well as the reliability. Fossil fuel, nuclear and hydroelectric plants were mainly utilized to generate power as they were the most economical and efficient at that time. Power consumption started increasing with time and it reached its peak in the 20th Century with the evolution of the industry and residential sector growth. To handle this drastic increase in energy demands, more electric power grids were needed to supply power during peak time and to protect the system from poor quality of power supply and voltage oscillations. Therefore, in order to cope with this issue, electricity provider introduced the DSM for improving the energy at consumption side through policies and various consumption modes of energy such as valley filling, load shifting, peak clipping and so on [26][27]. Customer demand curve comprises of valleys and peaks. Deterioration of peak demand decreases customer demand charges. Peak clipping is accomplished by explicit monitoring of the systems responsible for the peaks. It helps to balance the available capacity with demand without the need for extra generation, thus minimizing capital costs, maintenance, and fuel charges. Whereas Load Shifting allows consumers to shift load to off-peak hours. Fig 2 shows various DSM techniques.

In the 21st Century, due to the advancements in Information and Communication Technologies (ICTs), the power industry experimented integration of smart sensors to the power grid. Thus, the idea of SG became a reality. New features enabled adoptive billing mechanism, which led towards eliminating the requirement for precise measurement. Furthermore, advancements in renewable energy technology like solar, wind, tidal and geothermal energy not only enhanced the power generation capacity of the system but also eliminated the environmental harms unlike old modes of energy generation. Additionally, the concept and availability of different types of decentralized generation have contributed not only in supplying power but the cost of power distribution has also been reduced. Finally, due to advancements in communication along with other new technologies has made the system more efficient and reliable [12].

Several fields were positively affected by the evolution of power grids. From an infrastructure perspective, the use of an integrated communication network with SMs lead to the development of data centers for transmitting and storing the details of various apps or interfaces data, whereas, the consumer side, i.e., homes and industries became integrated to

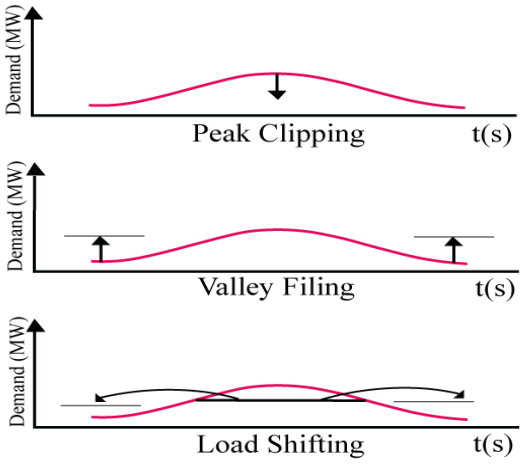


FIGURE 2. Demand Side Management Techniques.

the Microgrids. By the usage of various energy monitoring devices along with control sensors, e.g., lighting sensor, increased the energy efficiency and added extra comfort to the smart environment [28].

In smart facilities, energy consumption is measured and monitored by several devices that interact with a data center for providing gathered information accessibility to both customers and suppliers. This enables user to not only track their consumption of energy but can also automate various appliances if they want to maintain their energy consumption amenable. In contrast, energy suppliers can balance the grid's load by diverting the power to the facilities/locations where required, in particular where distributed energy resources (DER) are available [29][30]. Altogether, such smart systems lead to the birth of SEMS.

B. EVOLUTION OF SMART METERS

This section provides the evolution of meters leading to SMs as shown in Figure 3.

1) ELECTRO-MECHANICAL METERS

The first type and the most common among all types of electrical meters is the electromechanical watt-hour meter. It includes two inductive coils and a rotating metal disc. The inductive coils produce magnetic flux as power flows through them which further results in rotation of the disc. The rotation speed of the disc is proportional to the power and to bill the consumers for their energy consumption, the disk revolutions are counted. The simplicity in the design made electromechanical meters popular, although they did not provide any additional features. With technological advancement, the power grids required improved monitoring and controlling aspects of energy meters which lead to the development of electronics meters [31].

2) ELECTRONIC DIGITAL METER

Electronic meters are built on digital micro technology (DMT), sparing them from formerly used heavy moving components in the electromechanical meters. These meters are more accurate because of digital nature and laid the foundation

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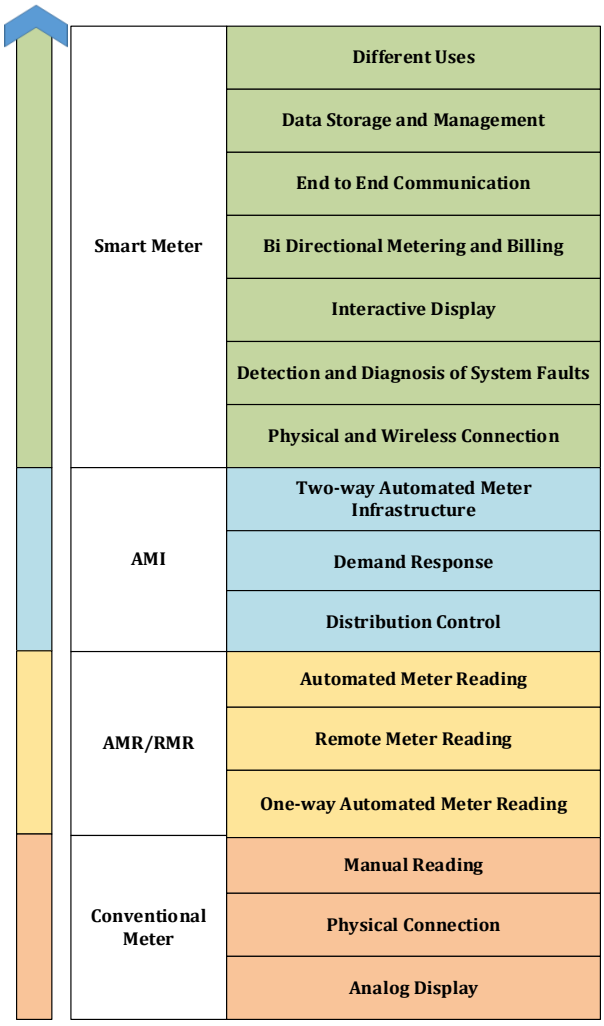


FIGURE 3. Functionalities and Development evolution of Energy Meters.

for SMs. As the technology progressed, extra features were introduced to make them smarter offering advantages for both consumers and suppliers. Furthermore, suppliers were able to deliver power efficiently by deploying load levelling and Real-Time Pricing (RTP) by using the remote access functions [32]. The evolution of electronic digital meters also gave flexibility as new features could be added by just plugging in extra modules [33].

The first-generation of digital electronics meters had the ability to report energy consumption to the supplier, thus achieving the concept of automated meter reading (AMR) and remote meter reading (RMR). Hence, energy suppliers could know about the energy consumption data without human intervention from long-distances [34]. This further led to the second generation of digital electronics meters incorporating advanced metering infrastructure (AMI) that could manage the energy demand [35].

AMI consists of smart energy consumption tracking which involves transmission and storage of the data using information exchange servers in data networks [36]. This lead to the development of SMs that made the concept of SGs a reality introducing the smart features like monitoring and controlling of the devices, bidirectional communication and billing, data storage and load management, identification of system's faults and power thefts, the evolution of smart cities, security and safety improvement, and emission control [24].

The literature in [37] demonstrates that by analyzing the appliances usage status, estimates of the consumers' behavior can be determined without the support of refined algorithms or tools. SMs are discussed in the next subsection.

3) SMART METERS

SMs record power usage and transmit data to the central data server periodically for the purpose of monitoring, control, and analysis which leads to the development of effective and efficient SEMS [38]. Functioning of SMs is given in Figure 4 and is explained as follows.

As aforementioned in the introduction section, SMs, unlike normal meters, are not just limited to measuring energy consumption but they can also measure a wide range of electrical variables for example current, voltage, frequency, power and power factor [39]. These variables are recorded at a specific time and the granularity of recording of the data differs from few minutes to few hours depending upon the requirements. The frequent data recording provides a comprehensive insight of the events and allows to gain a better understanding of the load usage patterns. Load predictions often lead to poor DSM but thanks to the SMs since they can plan effective, efficient and reliable DSM strategies [40]. Besides, SMs initiate alerts and notifications in case of any electrical parameter deviating from the standard limit. These alerts are helpful in the identification and elimination of system deficiencies and power losses leading to efficient load management, fault analysis and load profiling.

The SM data is usually sent to the central server after periodic intervals where it can be accessed. The communication infrastructure makes it possible to share data between the SMs and the server using wireless or wired connection depending upon the distance from the server to utility [23]. There may be primary and secondary combination of links to exchange data that provides reliability and robustness in the communication system [24]. In general, data communication in SM enables real-time tracking and controlling of various electrical loads using bi-directional communication.

Furthermore, SMs also have the capacity to store data which is helpful in case of communication failure. Later, once the communication is reestablished, the stored data can be communicated to the server. Data storage capacity can be enhanced as per the granularity of the data, typically data is stored for about 2-3 months. Data storage provides protection against natural failures, especially meter tempering, that aims to undermine the communication infrastructure [47]. The SM data is also accessible via a web hosting solution or data collection program developed using application programming interface (API), which allows data analysis, monitoring, and

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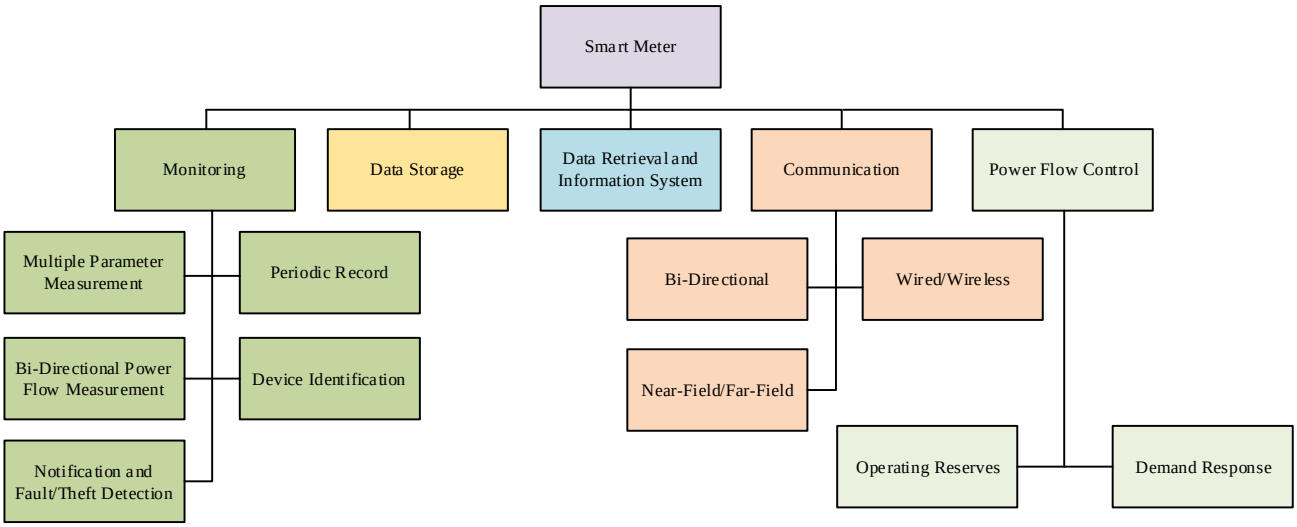


FIGURE 4. Functionalities of typical Smart Meters.

reporting. The data communication between the Web and SEMS is performed using a Middleware module [48] [49].

The control of power flow in SMs is allowed by demand response (DR) or by controlling the load at the consumer side. Non-critical loads can be disconnected during peak hours which further enables electricity suppliers to fulfil the peak demand [50] [51]. Customers housing an energy generating source and supply extra generating power to suppliers, need to keep track of their portfolio of consumption and generation. These customers can use SMs to accomplish this objective. Table 1 summarizes different communication technologies utilized in smart meters [41– 46].

III. THE SYSTEM MODEL

All the above functionalities are incorporated into a sole solution proposed in this work that is simple to install and scalable to implement. This work combines the contributions of the most promising solutions mentioned in [52].

TABLE I
COMMUNICATION TECHNOLOGIES OF SMART METER

Communication Technologies	Specifications
GPRS/GSM [41]	• Suitable for distantly located consumers. • Wide area coverage and low subscription cost
Zigbee [42]	• Suitable for Home Area Networks (HANs), automation of buildings and for hostile environments • Well Protected
WLAN/Wi-Fi [43]	• Suitable for NAN (Neighborhood area network) and HAN
Power line communication [44]	• No considerable infrastructure required • Attenuation of data & noise • Compatibility constraints
WiMAX [45]	• Suitable for wide range & for distribution area network (DAN) • Overpriced
Serial communication (Ethernet/RS-232) [46]	• Low cost and Reliable

A) OVERVIEW OF THE PRESENTED SYSTEM

Work performed in this paper provides wide-ranging load control and monitoring scheme for electrical and aggregated appliances, together with hardware capable of monitoring and evaluating grid efficiency parameters in real-time. External load control (e.g., air conditioning and lighting systems) is also available and the addition of the software framework allows the analysis of the grid parameters of deployed facilities by producing various management reports.

The hardware utilized to satisfy this proposal includes an intelligent SM designed to fulfill the technological characteristics mentioned in Figure 4. It is an affordable platform designed using MSP430F67641 micro controller unit (MCU) for there-phase energy monitoring and nodeMCU for developing bi-directional wireless interface with the Entrack based management platform. The design and architecture of the SM is discussed in the next subsection.

1) ARCHITECTURE OF THE SMART METER

The SM architecture includes three High-end micro-controller-system on Chips (SoCs) used in combination for transferring data to the IoT middleware module.

The first one used is Texas Instrument's MSP430F67641 MCU which is a powerful system on chip designed for grid parameters monitoring [53]. With a powerful structure it precisely computes voltage, current, power (active, reactive, and apparent), power factor and frequency. The mathematical model for the calculation of these parameters is provided in next subsection of this section. Specifications of this SoC module are given in Table II.

The second SoC used is Texas Instrument's TM4C129x which is a Tiva C series microcontroller equipped with an ARM-Cortex-M architecture [54]. This MCU is chosen to manage connectivity along with data storage in its non-volatile memory. This is also capable of controlling loads remotely

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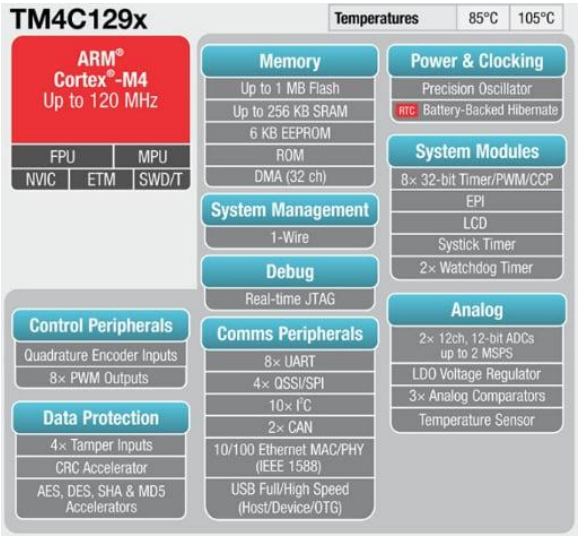


FIGURE 5. TM4C129x MCU Specifications.

TABLE II
CHARACTERISTICS OF MSP430F67641.

Processor	CPU	A/D Converter		I/O Pins	Power Ratings
16 Bit	25 MHz	Voltage	Current	72	3.3VDC
		10 bit	24 bit		

(such as on/off, cooling, and lightning circuits) along with real-time tracking of the key control variables like humidity, temperature, Photovoltaic (PV) sensors etc. (not included in scope of this work). Further, it is connected using a universal asynchronous receiver / transmitter (UART) to the third SoC, the nodeMCU microcontroller, which is an IoT platform [55].

NodeMCU uses the message queuing telemetry transport (MQTT) protocol for wirelessly communicating with the IoT middleware. In this work we have used IEEE 802.15.4 for wireless connection interfacing. Specifications of the TM4C129x MCU are shown in Figure 5 and for nodeMCU are shown in Table III.

2) MATHEMATICAL MODEL FOR GRID PARAMETERS

In this section, we provide the mathematical model for measuring and collecting grid parameters by the Texas Instrument's MSP430F67641 MCU platform, i.e., voltage, current, power (active, reactive, and apparent), power factor and frequency [56].

The analog input circuit given in Figure 6 (a) measures the phase current by Current Transformers (CTs), which supports internal currents up to 50 mA, and the input circuit given in Figure 6(b), measure the voltage, which computes voltage across the resistor burden of 12.40 Ω. The maximum allowed

TABLE III
CHARACTERISTICS OF NODEMCU [55].

Manufacturer	Type	CPU	Memory	Input	Power
Espressif Systems	32-bit MCU	160 MHz	4MB flash	48 pins	3.3 V DC

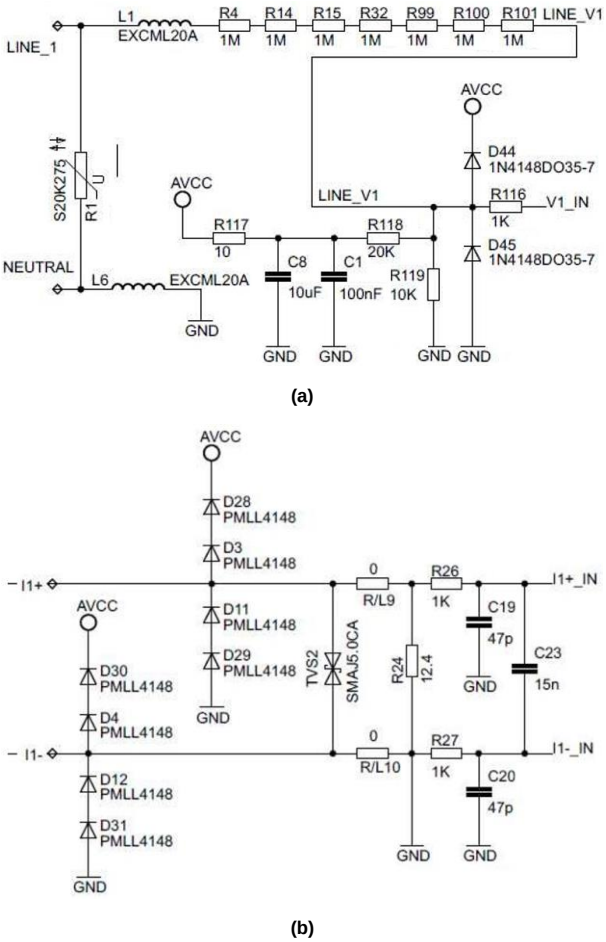


FIGURE 6. MSP430F67641 MCU Input circuits of current and voltage monitoring. (a) Input circuit for voltage monitoring. (b) Input circuit for current monitoring.

range of voltage computation is up to 270 V phase to neutral with surge suppressor offered by transient voltage suspension (TVS) diodes and varistors. Both circuits given in Figure 6 are front end analog circuits, so they require analog to digital converters (ADCs). The ADC voltage and current specifications along with MSP430F67641 details are shown in Table II.

Current and Voltage samples are attained from the ADC-24 and ADC-10 respectively at a sampling rate of 4096 Hz. Following equations are used for computing RMS values of voltage and current [56]:

$$V_{PhaseRMS} = K_{vPhase} \times \left(\sqrt{\frac{\sum_{n=1}^{Sample\ Count} v_{Phase}(n) \times v_{Phase}(n)}{Sample\ Count}} - v_{Phase\ Offset} \right) \quad (1)$$

$$I_{PhaseRMS} = K_{iPhase} \times \left(\sqrt{\frac{\sum_{n=1}^{Sample\ Count} i_{Phase}(n) \times i_{Phase}(n)}{Sample\ Count}} - i_{Phase\ Offset} \right) \quad (2)$$

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where,

- $v_{Phase}(n)$ = Sample of voltage at an instant n $v_{Phase\ Offset}$ = Offset used to subtract effects of the Additive White Gaussian Noise (AWGN) from the voltage converter
- $i_{Phase}(n)$ = Sample of current at an instant n
- $i_{Phase\ Offset}$ = Offset used to subtract effects of the AWGN from the current converter
- $Sample\ Count$ = Number of samples in specific time frame
- K_{vPhase} = Scaling factor of voltage
- K_{iPhase} = Scaling factor of current

Active and Reactive Powers are calculated using Equations (3) and (4).

$$P_{ACTph} = K_{ACTph} \times \frac{\sum_{n=1}^{Sample\ Count} v(n) \times i_{Phase}(n)}{Sample\ Count} \quad (3)$$

$$P_{REACTph} = K_{REACTph} \times \frac{\sum_{n=1}^{Sample\ Count} v_{90}(n) \times i_{Phase}(n)}{Sample\ Count} \quad (4)$$

where,

- K_{ACTph} = Scaling factor for active power
- $K_{REACTph}$ = Scaling factor for reactive power
- $v_{90}(n)$ = Voltage sample at a sample instant n shifted by 90 degrees
- The 90° phase shift technique is used for reactive power due to following reasons:
- Very small current is required to precisely quantify reactive power.
- Compliance with the internationally defined system of measurement is required.

Apparent power of each phase is calculated using Equation (5), whereas cumulative apparent, reactive, and active powers of each phase is calculated using Equations (6), (7) and (8).

$$P_{APP(ph)} = \sqrt{P_{APP(ph)}{}^2 + P_{REACT(ph)}{}^2} \quad (5)$$

$$P_{ACT(Cummulative)} = \sum_{ph=1}^{3} P_{ACT(ph)} \quad (6)$$

$$P_{REACT(Cummulative)} = \sum_{ph=1}^{3} P_{REACT(ph)} \quad (7)$$

$$P_{APP(Cummulative)} = \sum_{ph=1}^{3} P_{APP(ph)} \quad (8)$$

Power factor is determined after the computation of active and apparent power using the following equation.

$$Power\ factor = \frac{P_{ACTIVE}}{P_{Apparent}} \quad (9)$$

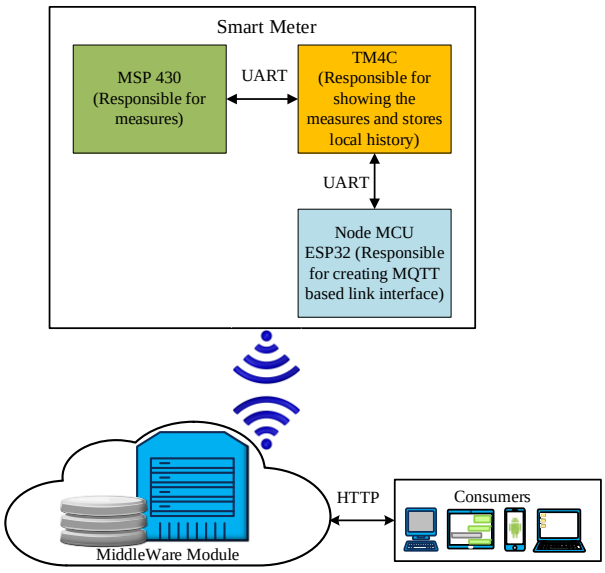


FIGURE 7. SMES Block diagram.

2) THE SMART ENERGY MANAGEMENT SYSTEM (SEMS) WORKING

In this section the working of SEMS using the above-described SM is detailed. The block diagram of the presented SEMS is given in Figure 7 and working described below is based on the flowchart given in Figure 8. The following steps indicate the functioning of the SEMS:

- The MSP430F67641 MCU reads the parameters of the grid, gathers data such as current and voltage and transmits the data via UART to other MCU TM4C129x.
- TM4C129x collects and stores the data and updates it with an interrupt in EEPROM memory roughly once in every 30 seconds.
- In the meantime, a function in TM4C129x setups a link with each collected grid parameter obtained through MSP430F67641 to a valid JavaScript Object Notation (JSON) format. This enables it to transmit the data to the third MCU, i.e, nodeMCU, also using UART. Which further empower the SM to facilitate MQTT protocol for IoT middleware module [57]. Here also, the data is roughly carried out in every 30 seconds.
- The data obtained from above mentioned process is transmitted to the IoT middleware that links basic systems like various IoT based devices to mediator applications. It functions as a translation layer that facilitates connectivity, data storage for specific inquiries, continuous availability of data to the users and processing in real time.
- As SEMS should support bidirectional communication, so, for remote and real-time accessibility of all the data, an API [58] is used to link the IoT middle to the consumers.
- Additionally, numerous reports are provided by the analytical engine server tool, that further enables

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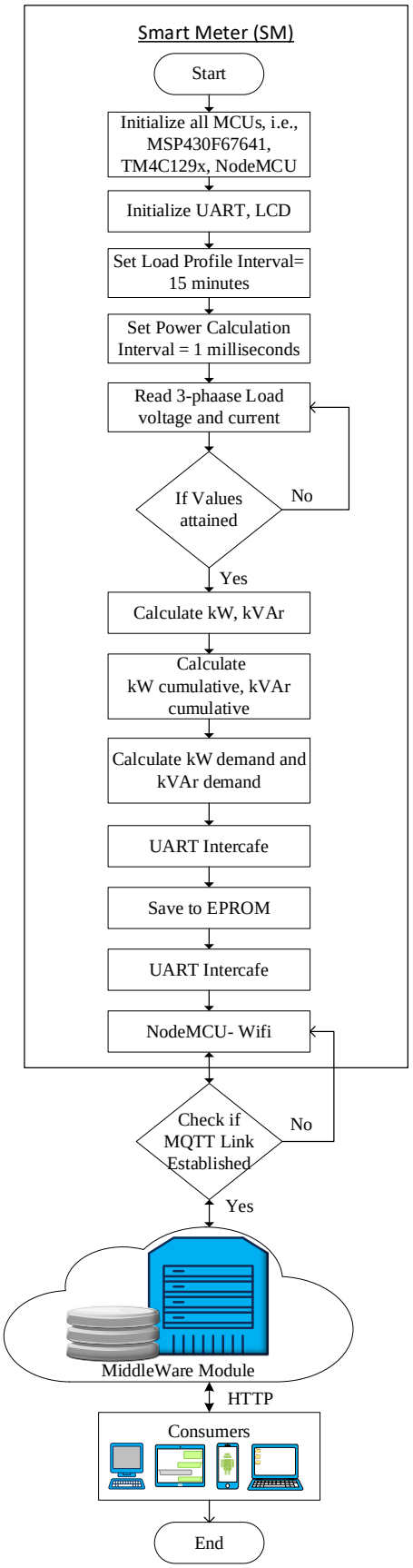


FIGURE 8. Operational Flowchart of SEMS.

SEMS to evaluate efficiency of the electricity provided to the facilities, as well as provides characteristics of usage and analysis of cycles at different times (instantaneous, daily, and monthly).

The next section provides the implementation and validation of the above-described SEMS.

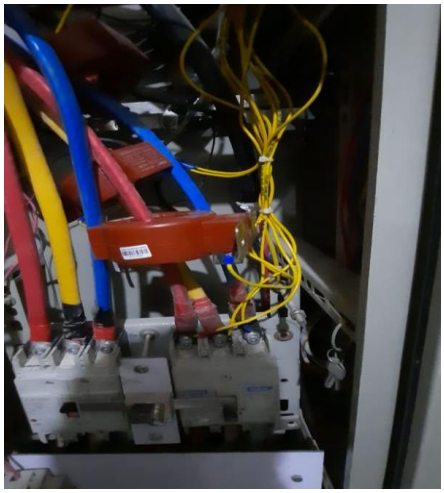
IV. DEMONSTRATION AND VALIDATION: A CASE STUDY BASED ANALYSIS

Demonstration and validation of the above-described SEMS is based on a case-study based analysis developed through testing and implementation of the above-described SEMS in four major locations of a well-known private company in Pakistan, i.e., Stylo Pvt. Ltd.

To test and evaluate the performance of presented SEMS framework, SMs along with 100A CTs were installed at the



(a)

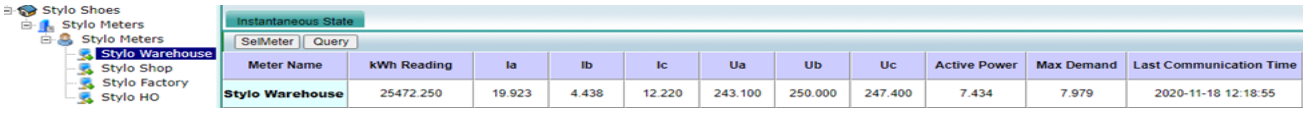


(b)

FIGURE 9. (a) Power distribution panel with Smart Meter. (b) Power Distribution Panel with CT's.

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Meter Name	kWh Reading	Ia	Ib	Ic	Ua	Ub	Uc	Active Power	Max Demand	Last Communication Time
Stylo Warehouse	25472.250	19.923	4.438	12.220	243.100	250.000	247.400	7.434	7.979	2020-11-18 12:18:55

(a)

Instantaneous Grid | Instantaneous Curve

30 minutes | PT/CT | V | I | MD | cosφ | kW | StartDate: 2020-11-18 00:00 | EndDate: 2020-11-18 19:00 | SelMeter | Query

No.	Terminal name	Meter name	Data date	Data time	Voltage			Current			Demand	Power factor	Power
					A phase	B phase	C phase	A Phase	B Phase	C Phase			
1	Stylo Warehouse	Stylo Warehouse	2020-11-18	00:00	265.200	267.600	265.400	12.740	3.885	15.150	6.423	99.800	6.517
2	Stylo Warehouse	Stylo Warehouse	2020-11-18	00:30	260.400	262.400	259.800	12.249	3.959	15.086	6.429	99.700	6.190
3	Stylo Warehouse	Stylo Warehouse	2020-11-18	01:00	262.900	264.000	262.400	12.464	3.979	16.277	6.346	99.800	6.514
4	Stylo Warehouse	Stylo Warehouse	2020-11-18	01:30	254.600	255.900	254.100	12.554	4.088	16.166	6.311	99.700	6.361
5	Stylo Warehouse	Stylo Warehouse	2020-11-18	02:00	253.300	257.500	254.000	13.855	4.450	16.854	6.455	99.400	6.398
6	Stylo Warehouse	Stylo Warehouse	2020-11-18	02:30	254.800	259.200	255.700	13.912	4.584	16.906	6.385	99.400	6.412
7	Stylo Warehouse	Stylo Warehouse	2020-11-18	03:00	254.900	259.400	255.900	13.987	4.556	17.004	6.396	99.500	6.513
8	Stylo Warehouse	Stylo Warehouse	2020-11-18	03:30	251.400	256.600	254.500	14.375	4.553	16.042	6.513	99.300	6.350
9	Stylo Warehouse	Stylo Warehouse	2020-11-18	04:00	254.200	258.600	256.100	13.885	4.587	15.984	6.394	99.100	6.177
10	Stylo Warehouse	Stylo Warehouse	2020-11-18	04:30	253.800	258.300	255.900	14.408	4.613	16.124	6.412	99.200	6.395
11	Stylo Warehouse	Stylo Warehouse	2020-11-18	05:00	250.500	255.500	253.400	13.812	4.560	16.007	6.509	99.100	6.135
12	Stylo Warehouse	Stylo Warehouse	2020-11-18	05:30	249.400	254.500	252.800	13.762	4.525	15.815	6.296	99.200	6.123
13	Stylo Warehouse	Stylo Warehouse	2020-11-18	06:00	244.800	250.700	247.900	13.783	4.433	15.815	6.143	99.100	6.063
14	Stylo Warehouse	Stylo Warehouse	2020-11-18	06:30	244.500	251.300	249.600	13.922	4.425	11.672	5.104	92.800	4.993
15	Stylo Warehouse	Stylo Warehouse	2020-11-18	07:00	242.000	248.100	245.200	13.928	4.381	12.441	5.099	92.800	5.103
16	Stylo Warehouse	Stylo Warehouse	2020-11-18	07:30	241.100	246.300	242.900	13.957	4.396	12.499	5.087	92.700	5.080
17	Stylo Warehouse	Stylo Warehouse	2020-11-18	08:00	240.500	247.600	244.900	14.688	4.396	11.755	5.697	95.900	5.582
18	Stylo Warehouse	Stylo Warehouse	2020-11-18	08:30	252.100	259.600	256.100	16.865	4.368	11.855	5.981	97.300	6.448
19	Stylo Warehouse	Stylo Warehouse	2020-11-18	09:00	244.600	247.400	249.600	17.254	4.436	12.363	5.998	97.400	6.340

Total check 155 results.

(b)

Sdate: 2020-09 | Edate: 2020-10 | SelMeter | Query

No.	Terminal name	Meter name	Date	Forward Energy indication		Forward Energy kWh	
				Active sum	Reactive sum	Active sum	Reactive sum
1	Stylo Warehouse	Stylo Warehouse	2020-09	14439.779	5669.560	9657.210	3896.951
2	Stylo Warehouse	Stylo Warehouse	2020-10	22935.650	8716.070	8495.871	3046.510
3	Stylo Shop	Stylo Shop	2020-09	183.800	35.759	13429.100	2640.900
4	Stylo Shop	Stylo Shop	2020-10	289.090	48.860	10529.000	1310.100
5	Stylo Factory	Stylo Factory	2020-09	739.070	160.750	22958.400	4976.840
6	Stylo Factory	Stylo Factory	2020-10	1253.640	278.060	20582.800	4692.400
7	Stylo HO	Stylo HQ	2020-09	459.240	183.060	58601.600	23305.760
8	Stylo HO	Stylo HQ	2020-10	780.490	302.720	51400.000	19145.600

(c)

Figure 10. Example API View (a) Instantaneous State of Smart Meter. (b) Instantaneous state for a specific time span. (c) Monthly kWh usage.

main distribution panels of four different facilities of Stylo Pvt. Ltd, at different locations in the city of Lahore, Pakistan. The details of facilities are provided in Table IV. The installed SM with 100A CT is shown in Figure 9. Each SM monitors the energy and power consumption of its respective location and forwards it to the Middleware module for further processing, i.e., the centralized cloud server. Furthermore, the real-time data collection, monitoring and accessing of various reports from the cloud server is made possible by accessing SM consumer application, i.e., Entrack software, as shown in Figure 10.

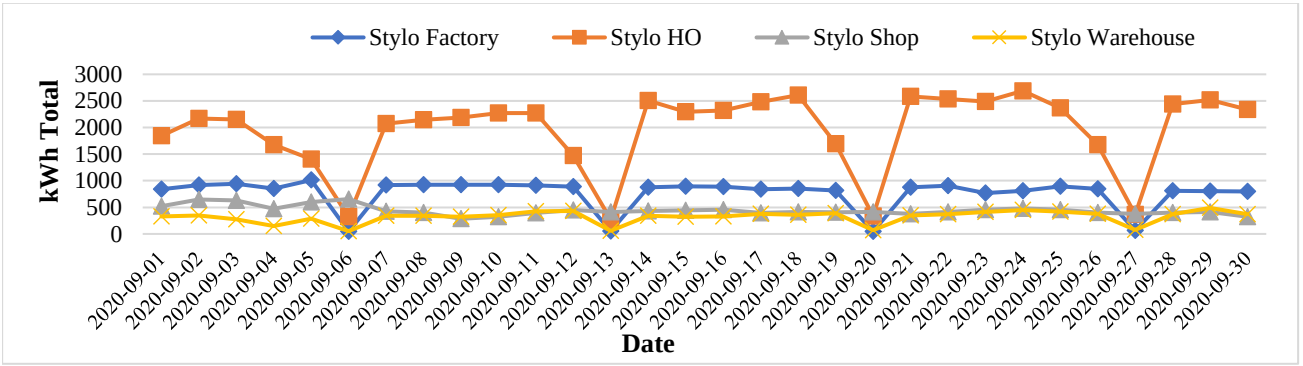
The SM collected data was evaluated using different timescales. Results in Figures 11 and 12 demonstrate and differentiate the gross use of energy and reactive power, respectively, for all four facilities from September 01, 2020 to November 17, 2020. It can be observed that the daily average power consumption of Stylo HQ in the months of September and October is around 2300 kWh, as shown in Figures 11a and 11b, which reduces to a daily average of around 1300 kWh, as shown in Figure 11c. Similar trend is observed for the reactive power results shown in Figure 12. This is because of reduction

TABLE IV
DETAIL OF FACILITIES WHERE SMART METERS INSTALLED

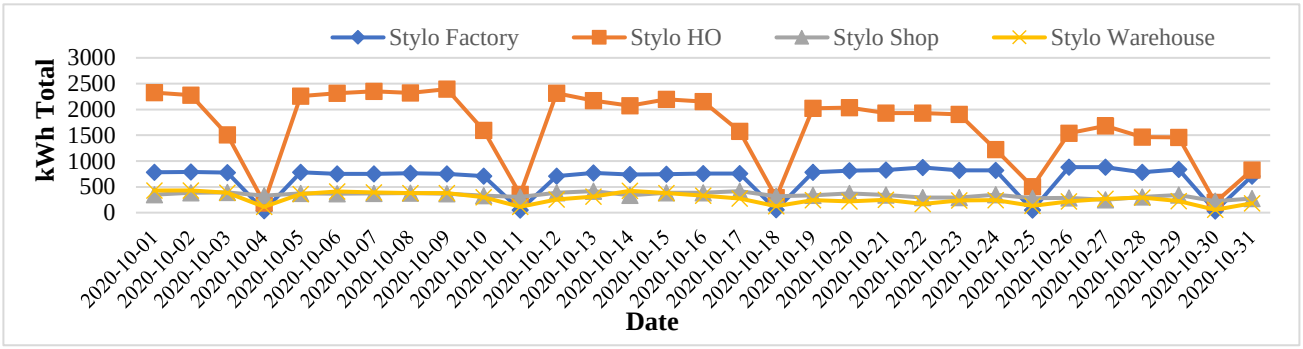
Sr. No	Smart Meter Name	Facilities Type	Working Days	Timings	
				Working	Break
1	Stylo HQ (Head Quarter)	Corporate Head Office	Monday to Friday	9 AM to 6 PM	1 PM to 2 PM
		Pret Production and Shoe Finishing Lines (Industry)	Monday to Saturday	8:30 AM to 5:30 PM	12:30 PM to 01:30 PM
2	Stylo Factory	Production Factory (Industry)	Monday to Saturday	8:30 AM to 5:30 PM	12:30 PM to 01:30 PM
3	Stylo WH (Warehouse)	Shoe Warehouse	Monday to Saturday	9 AM to 9 PM	1 PM to 2 PM
4	Stylo Shop	Shoe Outlet (Commercial)	Monday to Sunday	11 AM to 11 PM	No Break

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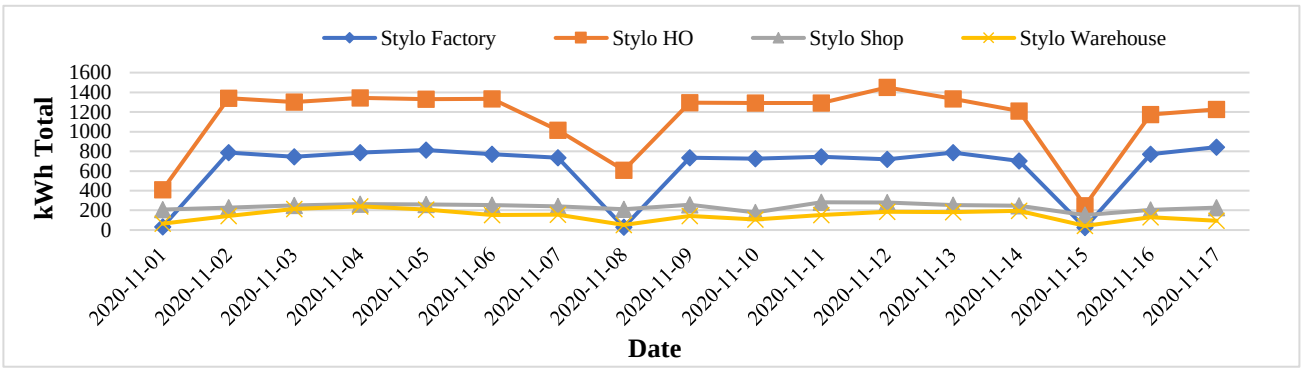
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(a)



(b)



(c)

FIGURE 11. (a) Daily Gross use of energy for the month of September 2020. (b) Daily Gross use of energy for the month of October 2020. (c) Daily Gross use of energy for the first 17 days of November 2020.

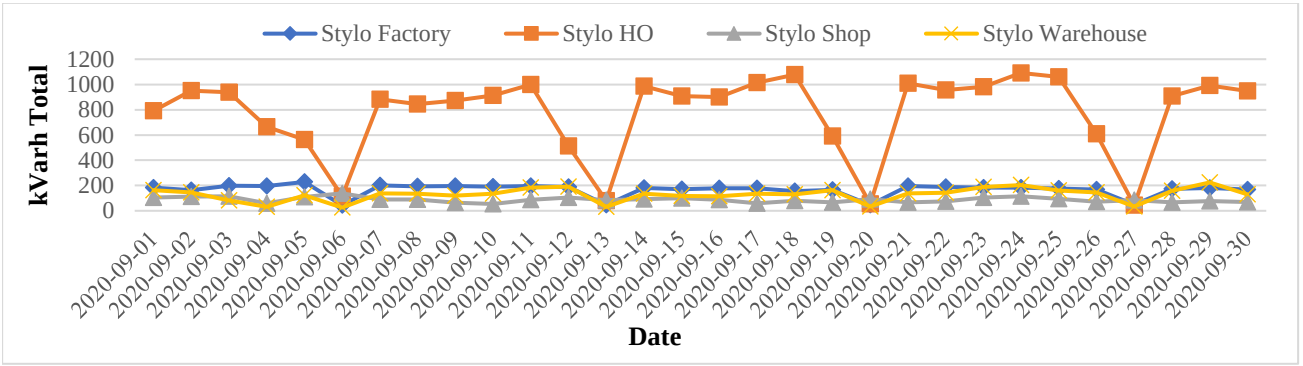
in the usage of central air conditioning units due to climatic changes. Furthermore, the results of remaining facilities are much stable, this is because of the limited number of installed air conditioners (ACs). Results for the breakdown of weekdays-based net average energy consumption, derived from the results in Figure 11, are given in Figure 13.

From these results, it can also be observed that the average electricity consumption of Stylo HQ drops by 20% on Saturdays, since it is considered as an off day for the corporate office. The electricity consumption percentage is almost the same for Stylo Shop as it remains open for 7 days a week. For Stylo factory and Stylo-Warehouse, electricity consumption is almost the same in working days whereas it drops to 10 % on Sundays because it is considered as a public holiday. This data also gives an insightful interpretation of the people's behavior working in the facilities, e.g., as if someone uses the facility in off days, then it can be monitored and sorted out accordingly. These analyses are very helpful for one to get the idea of when and where most of power is being consumed.

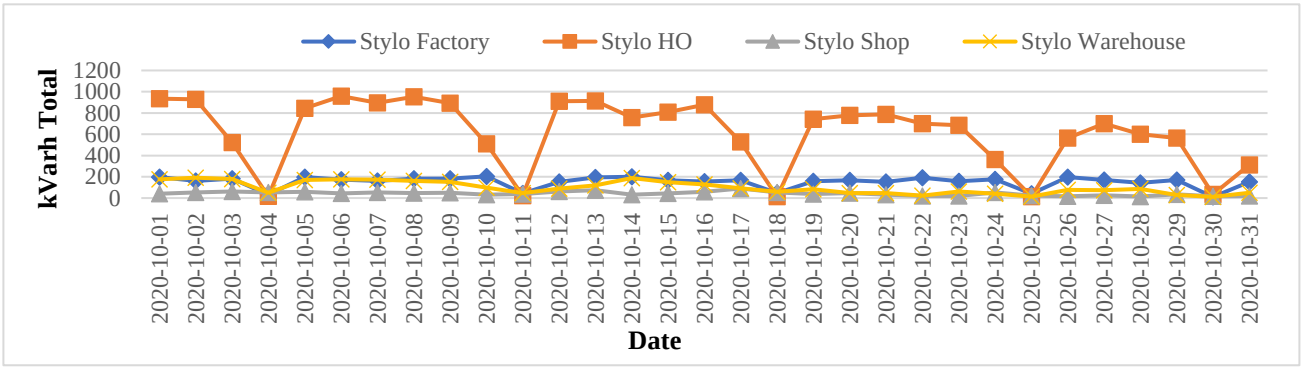
The immediate / instantaneous energy consumption data is also extremely critical as it may suggest potential problems inside an equipment or the overall facility. This kind of data can be effectively composed in the form of load consumption graphs. This approach would not only improve the system's effectiveness, but also dramatically decrease the maintenance and operational cost. For example, on a typical day of November 05, 2020, instantaneous electricity demand of 30 minutes' time duration is shown in figure 14(a). It is observed

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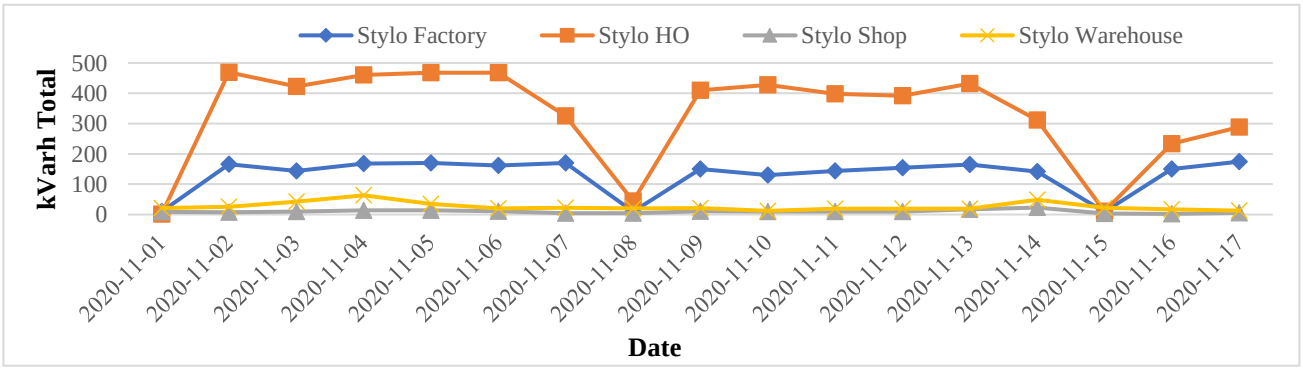
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(a)



(b)



(c)

FIGURE 12. (a) KVarh for the month of September 2020. (b) KVarh for the month of October 2020. (c) KVarh for first 17 days of November 2020.

that instantaneous demand is minimum in break hours due to the shutdown of various loads, especially AC units. From this data, we can also predict the base and the peak load for all facilities. Whereas half hourly maximum demand of Week 45 (November 01, 2020 to November 07, 2020) of all facilities is shown in figure 14(b). Figure 15 shows the average power factor of all facilities in Week 45, this kind of data can be helpful in load management.

V. POTENTIAL OF THE IMPLIMENTED SEMS

This analysis offers useful lessons that can support future researchers and developers to create or improve IoT-based energy management solutions.

The dilemma of unconsciously exploiting these resources is not overcome by an energy management solution itself but consumers must be prudent as well as conscious about the environment to discourage misuse of energy resources. The primary advantage is that a customer can get a service in which data on power usage can be received in real-time and can communicate with the installed loads (such as HVAC and lights). Furthermore, the presented SEMS in this work can be integrated into new or existing systems without the use of a copyrighted solution provided by any manufacturer. Therefore, IoT-based SEMS can be used in a huge variety of situations where quality control and energy usage are needed.

Various applications can be incorporated in the presented SEMS such as real-time monitor and control of power quality

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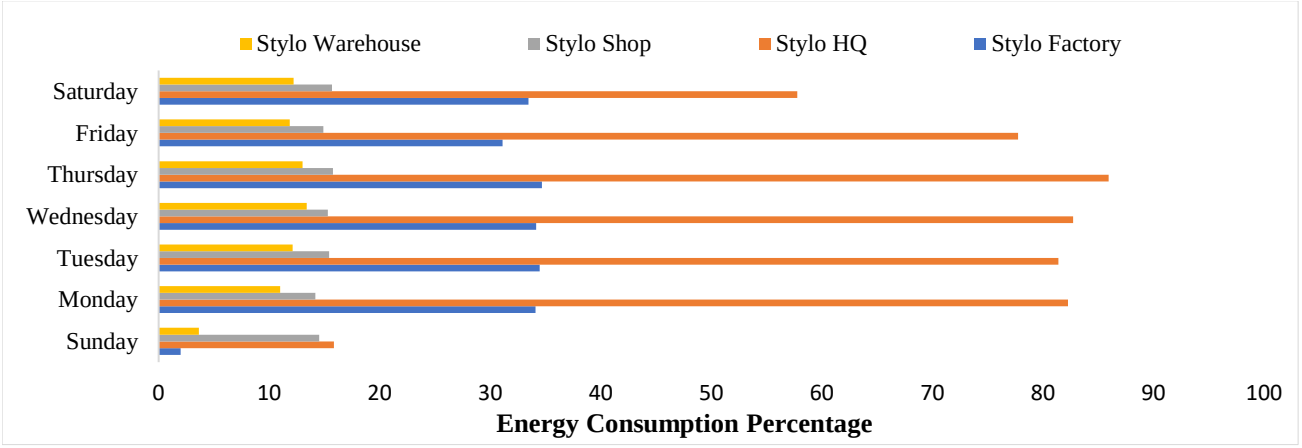
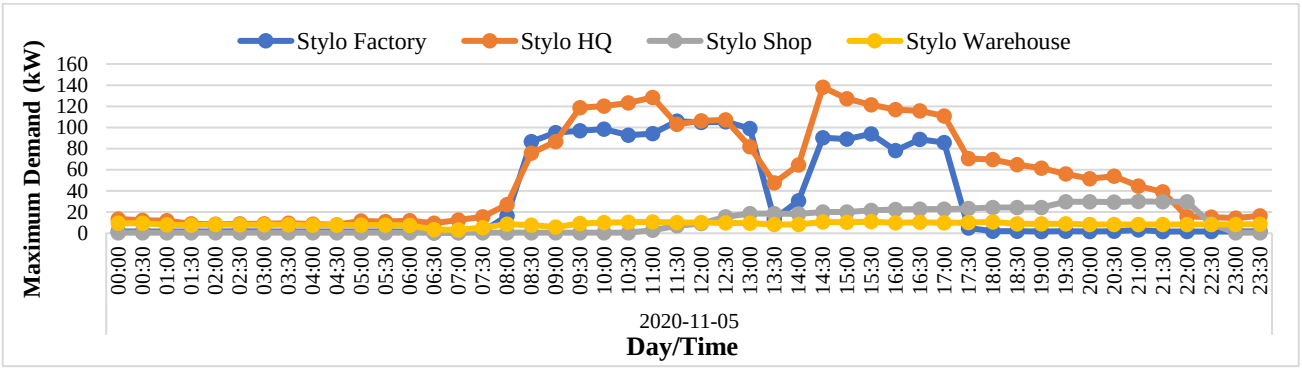


FIGURE 13. **Breakdown of net consumption based on weekdays.**

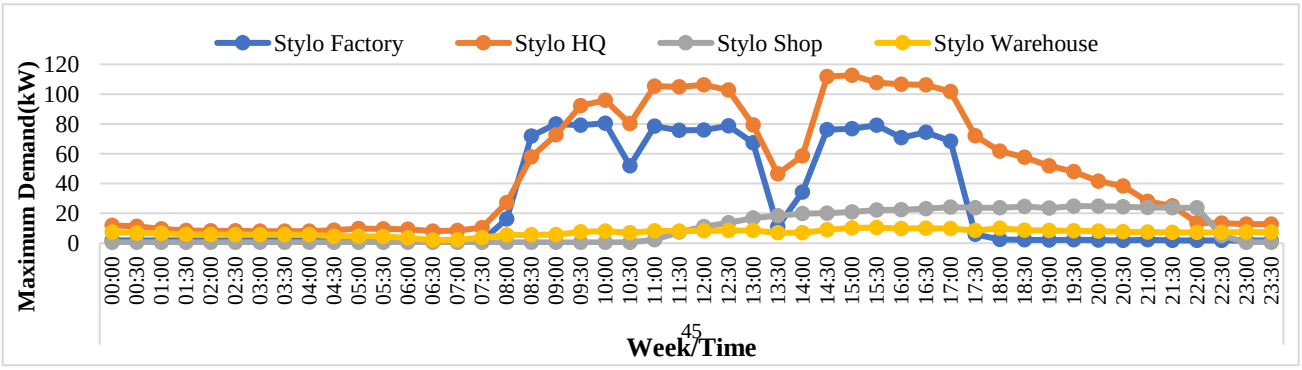
and energy usage. Consumers, with the help of a middleware module, can enable/disable HVAC or lighting circuits for efficient power consumption. This will prevent the depletion of energy resources. Such smart solution, if adopted by mass industries, can help identify energy-intensive equipment and processes enabling them to develop energy savings strategies, eventually resulting in overall power consumption of a larger region (a city or a country).

Fluctuations in the voltage and current often cause damages to industrial equipment which are very costly to repair or to replace. Using SMs on power lines within the industrial facilities would make it possible to track the efficiency of the power supplied by the utility. This tracked data can be used to create smart decision systems leading to the protection of heavy industrial equipment through early detection of power quality fluctuations and faults.

Energy auditing is the method of evaluating a system's energy efficiency and finding opportunities to enhance it. To achieve peak load reduction and cost optimization targets for industry, utility, and household facilities, energy audits play a significant role. These audits can be more productive when accurate demand profiling, hourly load diagrams, detailed user behavior and consumption breakdown insights of facilities are well known. SEMSs offer an easy and effective way of



(a)



(b)

FIGURE 14. **(a) Half hourly Maximum Demand (Nov, 05 2020). (b) Weekly Average Maximum Demand (Week 45).**

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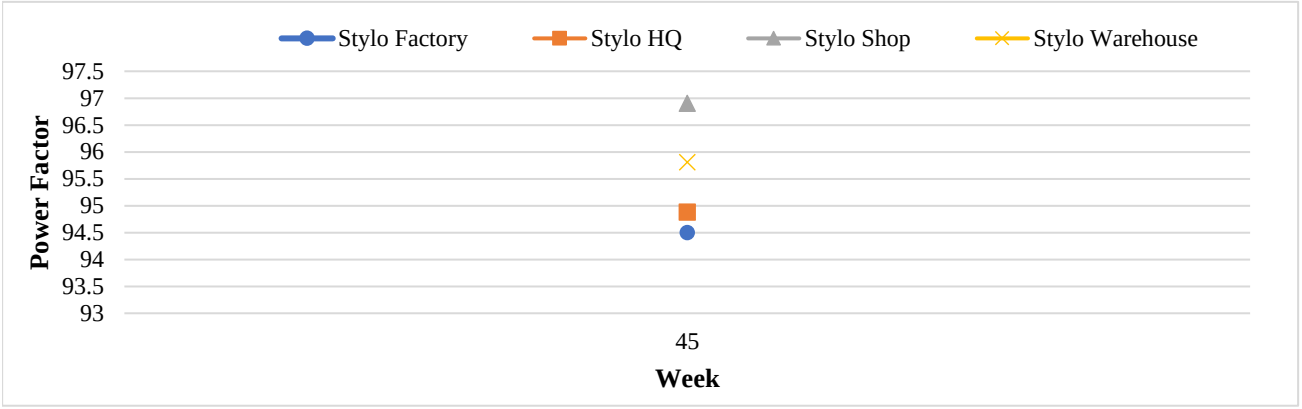


FIGURE 15. **Average Power Factor of all facilities for Week 45.**

conducting energy audits, thus enabling fruitful results. DR is a perfect example of energy auditing advantage. It is a response generated when an unusual change occurs in energy usage trend of a customer. The role of SEMS in DR event is presented in following steps:

- Step 1: Overload: The peak load of a system goes beyond a predefined threshold.
- Step 2: Notification: Overload notification is trigged by the SM at the server
- Step 3: Client response: The Server takes different actions upon client input, like disabling the unessential loads etc.

The major advantages of the SEMS are shown in Figure 16.

VI. CONCLUSION AND FURTHER DIRECTIONS

Main goal of this research is to develop, plan, construct, demonstrate, and validate a low-priced SEMS solution to track daily electricity usage using an IoT-based technique. Using IoT communication protocols, it collects and further transmits the gathered data to the middleware module, which manages and provides users with their electricity usage information via consumer application. For real time data collection, the installed SMs operate online. The presented system can be installed in smart homes or in other smart environments which require daily electricity consumption.



FIGURE 16. **Advantages of using smart meters as a part of SEMS.**

Normally, people monitor their electricity usage manually or depend upon the measurements taken by electricity service providing companies, which are usually erroneous or have deficiencies. Due to technological advance, consumers are now more educated and are much concerned about monitoring of their actual amount of electricity consumption and utilization. With the use of presented SEMS, consumers can monitor and conserve their energy consumption efficiently as now they can monitor their real time data on energy usage. This will eventually minimize their power consumption, thus saving both money and energy. This has been validated in this work through a thorough assessment of the presented SEMS using an extensive case study-based analysis.

In this work, it has been proven that IoT is an essential part of the presented SEMS, as it not only provides motoring of the real time data, but two-way control of the connected devices for efficient power management, both at the consumer and service provider ends. In short, using IoT gives an intelligent solution for daily based power monitoring and control. Based on the case study provided in this work, it is concluded that the presented SEMS helps consumers easily, effectively, reliably, and accurately track their power consumption, helping them to understand and control the unwanted use of power, hence, resulting in energy conservation. Furthermore, this work justifies the use of emerging information and communication technology (ICT) needs. The scope of this work was to provide the monitoring part of the SEMS and is now being extended for efficient control of the connected devices by the internet and furthermore, the extended work will also show that how the presented SEMS can help for real-time configuration and monitoring of smart power grids. For example, to give an idea, IoT provides various possibilities to exchange gathered data over the grid to boost its performance. This data can be transmitted to remote locations via SM's thus enabling utilities to control the power grids as per customer's needs.

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In future SEMS, IoT-based smart energy meters will be able to share usage as well as energy quality with power grids thus enabling them to improve their performance and billing. To enhance the SEMS further, the consumer side data measurements and their power consumption behaviors can be combined with machine learning and deep learning techniques to predict efficient power grid configurations for efficient power distribution. Nevertheless, there is a lot of room for improvements due to limitations, such as standardization of communication protocols, redesign of safety rules, plug-and-play assistance, inbound big data control, recalculation of charging metrics, and improvement of the existing power grid infrastructure especially in developing countries.

Finally, smart devices play a vital part in the scenario by offering surveillance, tracking, management, and remote control of SEMS. The consumers can also upgrade their energy conservation policies using online authentication systems and communicate them remotely to the policy server, which removes the constraint of using manual physical services. Last but not the least another significant point to be considered is how the utility companies bill the consumers and care for easy options to pay the services such as the use of cryptocurrency.

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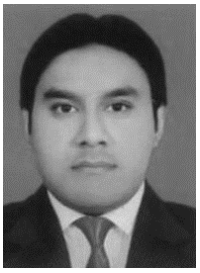
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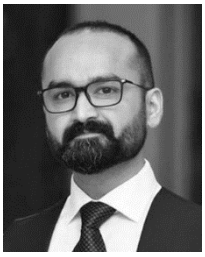
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